BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics (Objective Type) — Inter. (Part-I)-A-2021 (For all sessions) — Paper Code: 6191

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6191

Note: NOTE: Write answers to the questions on objective answer sheet provided. Four possible answers A,B,C & D to each question are given. Which answer you consider correct, fill the corresponding circle A,B,C or D given in front of each question with Marker or pen ink on the answer sheet provided.

Q1. Multiplicative identity of complex number is:

(A) (0,0)
(B) (0,1)
(C) (1,0)
(D) (1,1)

Q2. The contrapositive of $\sim p \rightarrow \sim q$ is:

(A) $p \rightarrow q$
(B) $q \rightarrow p$
(C) $\sim q \rightarrow \sim p$
(D) $\sim q \rightarrow p$

Q3. If A and B are any two non singular matrices then $(AB)^{-1}$ =

(A) $A^{-1}B^{-1}$
(B) $B^{-1}A^{-1}$
(C) BA
(D) AB

Q4. For a non-singular matrix A if XA=B then X =

(A) $A^{-1}B$
(B) BA^{-1}
(C) $(AB)^{-1}$
(D) $(BA)^{-1}$

Q5. If $f(x) = 3x^4 + 4x^3 + x - 5$ is divided by x+1, then remainder is:

(A) -6
(B) 7
(C) 6
(D) -7

Q6. If w is cue [sic] root of unity, then w^{15} =

(A) 1 (B) 0

(C) w (D) -w

Q7. Partial fraction of $(3x-11)/((x^2+1)(x+3))$ will be of the form.

(A) $(Ax+B)/(x^2+1) + C/(x+3)$ (B) $A/(x^2+1) + (Bx+C)/(x+3)$

(C) $(Ax+B)/(x+3) + C/(x^2+1)$ (D) $A/(x^2+1) + B/(x+3)$

Q8. If $a_n = (-1)^{n+1}$, then 26th term is:

(A) 1 (B) -1

(C) 26 (D) -26

Q9. $(n+1)^{th}$ term of G.P is:

(A) a_1r^{n-1} (B) a_1r^{n+1}

(C) a_1r^{n+2} (D) a_1r^n

Q10. n^{th} term of A.P is:

(A) $a_1(n-1)d$ (B) $a_1+(n+1)d$

(C) $2a_1+(n-1)d$ (D) $a_1+(2n-1)d$

Q11. With usual notation ${}^nC_r + {}^nC_{r-1}$ =

(A) ${}^nC_{r+1}$ (B) ${}^{n+1}C_r$

(C) ${}^{n-1}C_{r-1}$ (D) ${}^{n+1}C_{r-1}$

Q12. In the expansion of $(a+b)^7$, the second term is:

(A) a^7 (B) $7a^6b$

(C) $7ab^6$ (D) 8

Q13. In one hour, the hour hand of a clock turns through an angle.

(A) π/8 (B) π/4

(C) π/6 (D) π/2

Q14. 3π/4 radian is equal to:

(A) 110^0 (B) 135^0

(C) 150^0 (D) 130^0

Q15. $\sin(-300^0)$ =

(A) -√3/2 (B) √3/2

(C) 1/2 (D) 0

Q16. Period of sin x is:

(A) π (B) 2π

(C) 3π (D) -π

Q17. Radius of escribed circle opposite to vertex C is:

(A) Δ/(s-a) (B) Δ/(s-b)

(C) Δ/(s-c) (D) Δ/s

Q18. With usual notation a+b-c =

(A) 2S (B) 2S-2C

(C) 2S-2b (D) 2S-c

Q19. $2\tan^{-1}A$ =

(A) $\tan^{-1} 2A/(1-A^2)$ (B) $\tan^{-1} 2A/(1+A^2)$

(C) $\tan^{-1} A/(1-A^2)$ (D) $\tan^{-1} A/(1+A^2)$

Q20. Solution of $\cot\theta = 1/\sqrt{3}$ in quadrant III is:

(A) 5π/3 (B) 7π/6

(C) 4π/3 (D) 7π/3

Mathematics (Essay Type) — Inter. (Part-I)-A-2021 (For all sessions)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Q. Section - I

Q2. Write short answers of any eight parts from the following.

2x8=16

(i) Seperate [sic] into real and imaginary parts (2-7i)/(4+5i).

(ii) Factorize $3x^2 + 3y^2$.

(iii) Simplify (2,6)(3,7).

(iv) Let A = {1,2,3,4}, Find the relation {(x,y)/x+y < 5} in A.

(v) Write the inverse and converse of $\sim p \rightarrow \sim q$

(vi) Find the value of x if |3 1 x; -1 3 4; x 1 0| = -30 (3x3 determinant)

(vii) Find the condition that one root of $x^2 + px + q = 0$ is multiplicative inverse of other.

(viii) Evaluate $(1+w+w^2)(1-w+w^2)$.

(ix) Solve the equation ax = b where a,b are the elements of a group G

(x) Discuss the nature of roots of the equation $2x^2 - 5x + 1 = 0$.

(xi) If A = [1 2; a b] and A^2 = [0 0; 0 0] then find the values of a and b.

(xii) If A and B are square matrices of the same order, then explain why in general $(A+B)(A-B) \neq A^2 - B^2$.

Q3. Write short answers of any eight parts from the following.

2x8=16

(i) Which term of the A.P, -2,4,10,.....is 148?

(ii) Insert three G.M's between 1 and 16.

(iii) Write in factorial form (n+1)(n)(n-1)/(3.2.1).

(iv) Find the value of n, when ${}^{n}P_4 : {}^{n-1}P_3 = 9:1$

(v) If 5 is the harmonic mean between 2 and b, find b.

(vi) Find the number of diagonals of a 6-sided figure.

(vii) Evaluate $\sqrt[3]{30}$ correct to two places of decimals.

(viii) Expand by binomial theorem $(\sqrt{(a/x)} - \sqrt{(x/a)})^3$.

(ix) Resolve into partial fractions (7x+25)/((x+3)(x+4)).

(x) Resolve into partial fractions without finding the constants $(9x-7)/((x^2+1)(x+3))$

(xi) If 1/a, 1/b and 1/c are in G.P, show that the common ratio is $\pm\sqrt{(a/c)}$.

(xii) Check whether, $1 + 1/2 + 1/4 + ... + 1/2^{n-1} = 2(1 - 1/2^n)$ is true for n = 1,2.

Q4. Write short answers of any nine parts from the following.

2x9=18

(i) Prove that $\sec^2\theta - \operatorname{cosec}^2\theta = \tan^2\theta - \cot^2\theta$.

(ii) Find the values of $\cos 105^o$ taking $(105^o = 45^o + 60^o)$.

(iii) Prove that (sin8x + sin2x)/(cos8x + cos2x) = tan(5x).

(iv) Find the period of tan(4x).

(v) Show that $\gamma = (s-c)\tan(\gamma/2)$.

(vi) In $\triangle ABC$ a=3, b=6 and $B=36^o20'$ Find "b". [sic — printed verbatim]

(vii) Find area of $\triangle ABC$ if a=18, b=24 and c=30.

(viii) Find the value of $\cos^{-1}(-1/2)$.

(ix) Solve the equation 1 + cos x = 0.

(x) Find the soln [sic] of equation sec x = -2 which lies in [0, 2π].

(xi) What is the circular measure of the angle between the hands of a watch at 4 'o' clock.

(xii) Find the values of remaining trigonometric functions when $\cos\theta = 9/41$ and the terminal arm of the angle is in quad Iv

(xiii) If α, β and γ are angles of a triangle ABC then prove that $\tan(\alpha+\beta) + \tan\gamma = 0$.

Q. Section - II — 10x3=30

Note: Attempt any three questions from the following.

Q5. (a) If A = [2 -1; 3 1] verify that $(A^{-1})^t = (A^t)^{-1}$.

(b) Solve the system of equations x + y = 5; 2/x + 3/y = 2.

Q6. (a) Resolve 1/((1-ax)(1-bx)(1-cx)) into partial fractions.

(b) For what value of n, $(a^n+b^n)/(a^{n-1}+b^{n-1})$ is the positive Geometric Meam [sic] (G.M) between a and b.

Q7. (a) Prove that ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.

(b) If x is so small that its cube and higher powers can be neglected then show that $\sqrt{((1+x)/(1-x))} \approx 1 + x + 1/2\ x^2$.

Q8. (a) Two cities A and B lie on the equator such that their longitudes are 45^oE and 25^oW respectively. Find the distance between two cities, taking radius of earth as 6400 kms.

(b) Show that $\cos(\alpha+\beta)\cos(\alpha-\beta) = \cos^2\alpha - \sin^2\beta = \cos^2\beta - \sin^2\alpha$.

Q9. (a) The sides of a triangle are x^2+x+1, $2x+1$ and x^2-1. Prove that the greatest angle of the triangle is 120°.

(b) Prove that $2\tan^{-1}(1/3) + \tan^{-1}(1/7) = \pi/4$.